A Credit Risk Assessment System for Financial Institutions Utilizing Deep Learning

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***Abstract*— According to the banking supervision and regulation of the worldwide, credit risk approximation of a bank/financial institute is not only confined within a product line or value chain anymore. Today's banking regulation is putting pressure on the banks to deliver the risk measure values by regarding the risk considerations more broadly than just product and trading desk level together with the changes in the market environment and customers' commitments. Hence, it is deprived of fixed parameters and wholly relies on different parameters and huge data documents. Given very fast progress in the field of data availability, generation, storage, and data processing, machine learning now occupies a strategic position in all areas of the banking business and technologies. The usage of machine learning is extremely appreciable in the construction of credit risk modelling and loss estimations. Risk practitioners are inclining their focus toward applying the advantage of deep learning for the nonlinear relationships of substantive variables and default risk.**

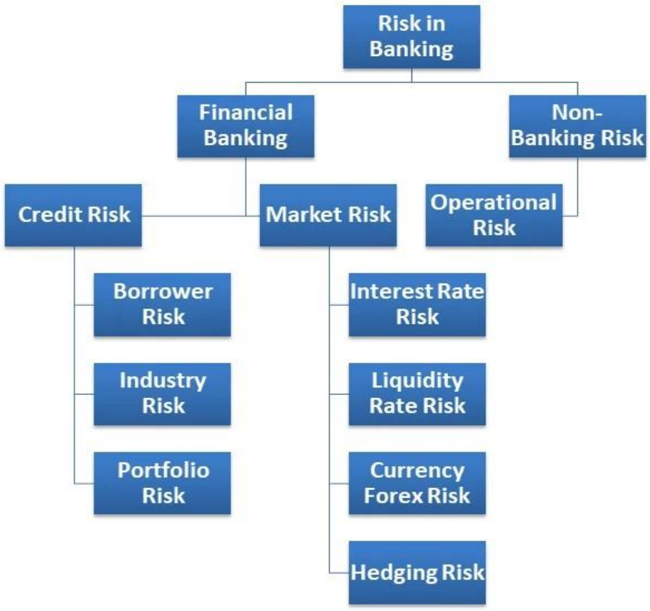
Keywords*— *credit risk, financial institute, customers, data processing, machine learning, feature engineering, deep learning, class imbalance.

I. INTRODUCTION

Any financial institution's primary function is to lend money to entities (borrowers) having deficit funds from entities (depositors) with excess funds. This is the fundamental idea behind capital allocation: the institute hopes to profit in some way from the process. The primary danger associated with this technique is that if the borrower defaults and fails to repay the funds, they will not be able to honour their debt. Credit risk is what this is, and it costs financial institutions money [1].

The entire risk landscape of a bank or other financial institution is shown above. The topic at hand belongs to the first credit risk category, which is consumer credit. Therefore, determining a customer's, legal entity's, or borrower's credit worthiness is crucial. Along with this risk, the bank's role in financial systems is subject to regulatory attention [2]. Types of banking risk is shown in Fig.1.

Even though a company's past default history does not guarantee that it will happen again, knowing the right amount



of capital is necessary before entering into any form of transaction. Three traits characterize credit risk at a very high degree.

Fig. 1. Types of Banking Risk [25]

- *Exposure At Default (EAD):* This describes the degree of exposure a bank has to a borrower entity. Simplistically said, it's the whole loan amount.
- P*robability of Default (PD):* This measures the chance that a borrower entity will default.
- Recovery Rate (RR) / Loss Given Default (LGD):

The amount (percentage) of the bank's invested funds that can be recovered (RR) in the event of a default. As a result, Loss Given Default (LGD) = 1-RR is found.

Based on the above, the expected loss (associated credit risk) can be calculated as based on Basel Committee on Banking Supervision 2005a, 2005b.

$$Expected\ Loss\ (EL) = Exposure\ at\ Default\ (EAD)$$
$$x\ Loss\ Given\ Default\ (LGD) x$$
$$Probability\ of\ Default\ (PD)$$

Banks actively monitor, control, and quantify these risks through risk management. The single biggest risk that banks encounter is credit risk [3]. Therefore, estimating the probability of default (PD) of a borrower entity is the most important metric to determine the credit risk of a deal with that business. Finding "Expected Loss (EL)" is not too difficult if PD is known. "Expected Loss" will show how much capital a bank needs to set aside in order to cover a deal's total loss in the event that an entity defaults.

Following and calculating PD from credit scores, such as S&P, Fitch, or Moody's ratings for corporate or government bonds, is one of the standard market activities. Another method is to treat the PD values of a security as if they were the same as the PD values of the security class. But when actual financial strain occurs, these strategies' flaws become apparent. Thus, a model-driven strategy using widely used statistical techniques developed. In this approach trade/transaction history and default events are used as input to the model [4]. Following statistical methods have been used for evolvingbetter models with high accuracy:

- Linear regression
- Logit and probit Models
- Neural networks
- Classification trees

Applying the statistical model through manual computations is laborious, inaccurate, and time-consuming. Keeping up with the rapid fluctuations in market data (prices, rates, etc.) and daily changes in client exposure would be exceedingly costly and challenging. As a result, the system has developed to apply statistical concepts to specific product and market exposure types.

But over time it has also proved as inefficient as holistic approach covers all products and all customers are needed to calculate the overall risk of an organization [5]. Both academic and industrial researchers are increasingly using machine learning (ML) and deep learning techniques that make use of these techniques. A summary of how machines and deep learning have been used to solve the problem may be found in the section below.

A. Description of Research Dataset

The dataset used in this study is a collection of financial statements presented in rows that was obtained from the Kaggle website. There are 778 characteristics and over 200,000 observations in the train data (used to identify the pattern). According to Kaggle, observations are unrelated to one another. Each observation mentions whether it is default or not default [6].

The measurable value of loss in the event of a default is mentioned. This number is in the range of 0 and 100. For instance, if 30 is regained, the loss is indicated as 70.

There is no loss if the value is 0. It is necessary to predict loss in the test set [7].

- *Data Description:* Columns
  - Description (Data Type)
  - Borrower ID
  - Feature 1
  - Feature 2
  - Feature 778
  - Loss: Indicator of loss/default (100) /Partial loss (between 0 to 100) / full payment (0) (Number)

B. Data Understanding and Quality Review

The first process that needs to be performed under the data understanding step is the assessment of the completeness and the quality of the provided dataset. In the given dataset, 20 features are of object data type, whereas 758 features are of number data type, and for almost all the features, missing values exist.

Since the features are anonymized because of the financial security, it again puts more constraints for a need to do an excellent quality check for the data like missing values, redundant information, skewed distributions, unexplainable features, outliers and so on, the noise always hurts the effectiveness of how the learning models are to be applied [8].

The pertinent literature that addresses the same or related problem categories is described in this section. A common worry for any bank or lending institution is loan default. Both from the standpoint of industrial practice and academia, attempts have been made to use machine learning and AI to address this problem in its entirety or a portion of it. There are two common methods which are [9].

- Supervised technique – Typically, this technique will be more effective when subject experts can of the data from its description meta data or statistical properties and guide the analysis in the right direction [10].
- Unsupervised technique – Here model gets trained on the permutations of the data given to it with no information about the labels to be assigned. It has become more useful if there is restricted possibility in analysis of data to get the strategy [11].

Structure of this study is going like that Section 2 is showing problem statement, Section 3 is showing AIMS and Objectives, Section 4 is showing research methodology, Section 5 is showing Discussion, and finally Section 6 is showing conclusion and future work.

II. PROBLEM STATEMENT

As part of its determinants, credit risk of a bank or lender is an essential commodity in the lending business. Learning more about the borrower in terms of credit risk, concerning his/her social, economic, academic and/or professional activity prepares the bank for the risk in the event of the bank having to deal with that individual as a borrower.

Hence deeper data analysis of default (full or partial) history of the customers from various socio-economic segments helps to determine the probability of default for "a potential customer" with reasonable certainty [12].

This subject has slowly shifted to being filled with large datasets in the last few years and there is pressure on efficient utilization of machine learning and deep learning to arrive at an optimized solution. The inherent effectiveness of the approach strongly relies on the most suitable quality data. The study, in this case, seeks to use the two models above in tandem so as to get the maximum results as a way of getting the solution to the probability. Someof the challenges of this study are outlined below: -

1) Non availability of Meta data: If the available dataset does not have any data description, no subject knowledge can be harnessed for usability or association, dependency,redundancy, importance of the features. Study will be reliant on statistical measures and the features to determine the above [13].

2) Class Imbalance: The unequal spread of data hinders an effective flow of learning as it brings an influence of bias. While using the classification, if one of the classes is not represented in its right proportion the algorithms do not identify the pattern for the said class.

3) Feature Engineering/ Feature Reduction/ Selection Techniques: To capture default history with product details, and customer's socio-economic details, each default / non-default instance are defined by lot of variables, where default occurrence is dependent variable and all other independent variables [14]. In the selection of features some sort of feature engineering or some kind of reduction of optimization is detrimental for choosing the best set for the computation of learning algorithm without leaving out any significant features for the computation of the learning algorithm. The feature reduction technique works by collapsing original features into new features or components with less figures of the data set. This technique's main emphasis is then on the correlation of the features where multicollinearity is kept off. For example:

- *Principal Component Analysis (PCA)*: It has been widely applied in data mining to investigate data structure. In PCA, new orthogonal variables (latent variables or principal components) are obtained by maximizing variance of the data. The number of latent variables (factors) is much lower than the number of original variables, so that the data can be visualized in a low- dimensional PC space [15]. Feature (variable) selection can be achieved either by choosing informative variables or discarding redundant variables.

- *Feature Selection (FS):* Feature selection technique reduces the number of features by selecting the subset of original features [16]. For example: A study has been done to find out effective features identified through Permutation Importance [17], where importance of selected features from the total features (243) were verified through prediction error while training the random forest model. And later study has been verified by subject matter experts.

- *Classification Metric:* The primary focus of default history data is on clients that fail, resulting in a variety of losses for the bank or financial institution. Therefore, it is evident that the main goal is to determine whether a prospective client might become untrustworthy, a circumstance that is represented in classification. This is justified by the fact that many variables together affect the outcome to categorize. Literature review and guidance plays a role in finding out effective metrics to determine the right measure [18].

III. AIMS AND OBJECTIVE

In the quest of finding the answers to above research questions, this proposed study will be around following sets of objectives: Identify and discover more about the deep learning processes on the given datasets and establish which shortcuts perform the best based on the responses of conventional machine learning algorithms. There is no description or meta data attached to the dataset used in this investigation.

It reduces the possibility or extent of comprehending the data's implications considering the loan default situation. For all areas of processing and learning the data, such as statistical understanding data, feature importance, and learning patterns, the study will therefore mostly rely on the evaluation of the statistical models' robustness.

A. Significance of the Study

In today's world it is essential that from the existing and past customer financial profile and credit behaviour one has to distinguish the credit records. This means if there was surety on how efficient the aspects captured would be in accurately identifying an individual then the system would be in a position to make a decision as to the extent to which an individual could be trusted in future sustainable financial engagements. This exercise is exhaustive as it considers various aspects and data points and requires careful selection of data attributes/features to establish a right predictive pattern utilizing both explicit and implicit information [19].

B. Scope of the Study

Using the dataset of Kaggle website (Loans' default prediction dataset is openly available at this link: (https://www.kaggle.com/competitions/loan-default-prediction /data), and the objective of the study would be to see how the problem can be tackled using Deep Learning like neural network-based model.

However, scope of study also comprises of to what extent deep learning is better than the Supervised Learning / ML techniques. Based on the previous discussions, another relevant consideration of the study is to compare the model performance of in with regards to successful feature engineering/feature reduction [20]. Overview of core concept of traditional feature reduction technique is shown in Fig.2.

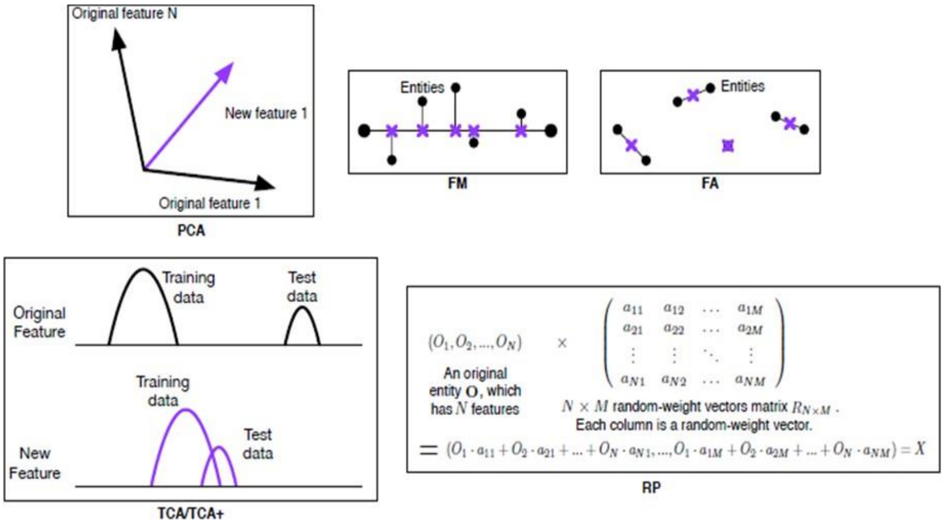


Fig. 2. Overview of core concept of traditional feature reduction technique

C. Class Imbalance and Data Size Change for Class Balancing

Even though loan default is one of the lender's most important concerns, the incidence of default for any given client base or product category is always much lower than the non-default rate. Therefore, the number of default observations in any collection of realistic market data will be quite small in comparison to the population of observations pertaining to non-defaults.

Class disparity results from this. Gradient boosting and random forest classifiers are equipped to overcome class imbalance problem. But for other classification models, it is importantto apply some data augmentation techniques to balance the distribution [21]. Data size change for class balancing is shown in Fig.3.

There are various techniques to do class balancing for the dataset under research with class imbalance. A few of the common methods are like reducing the size of majority classto the size of minority class or under sampling or increasing the size of minority class to the size of majority class orover-sampling or combination of both [22].

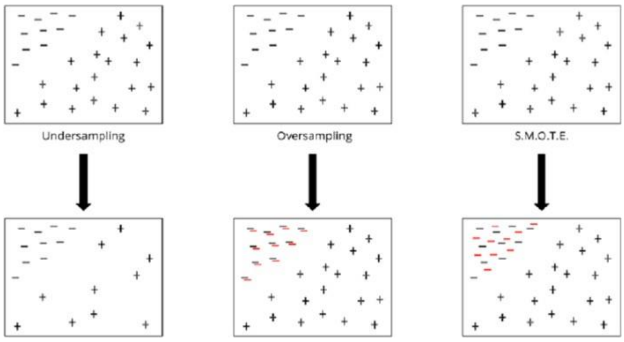


Fig. 3. Data size change for class balancing

There are drawbacks to both strategies. Random under-sampling (RUS) results in information loss, whereas Random over-sampling (ROS) causes model overfitting. Synthetic Minority Over-sampling Technique (SMOTE), or its

variants [23] are quite effective in doing over-sampling and appears to be most effective in most of the class imbalance problems.

IV. Proposed Methodology

The procedures listed below are carried out step-by-step starting with the pre-processing of the data, learning the data through models, and using the model-unknown test data to confirm accuracy:

- Review and comprehension of datasets.
- Feature engineering.
- Model training and implementation.
- Model assessment and testing.
- Work on potential enhancements (Hyper-parameter tuning)
- Result analysis
- Recording for further comparisons

The above steps are sequential and interdependent. An important point to mention is that the research dataset does not have the meta-data or data description associated with it. All the features are numeric with columns names are in ‘f (column number)’ format [24]. And the last column ‘Loss’ indicates the percentage of loss the bank has incurred (0 – for no loss, 100 – total loss and there are in-between percentage of losses too).

So, there is no scope to apply subject knowledge by understanding the implication of features through its data description [17]. Feature selection is very much dependent on the statistical evaluation of the impact of independent features (f1, f2, f3, ... f778) on the dependent feature (Loss). Proposed methodology is shown in fig.4.

V. MODEL BUILDING AND RESULT ANALYSIS

Well-known supervised machine learning classification models are trained in this work using the procedures listed below, and the outcomes are recorded.

- Logistic Regression
- Random Forest Classification

Additionally, the model's hyperparameter was adjusted to yield the best comparative results:

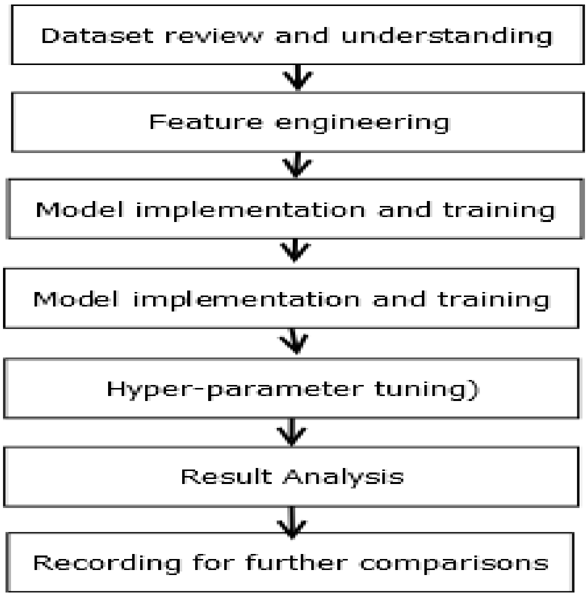


Fig. 4. Proposed Methodology

- Number of Estimator
- Depth of Tree
- Learning Rate
- Subsample

With the use of Scikit-learn's gridsearch cross validation, hyper-parameter tuning is accomplished by identifying the optimal combination of parameter values based on assessment criteria from a set of pre-defined possibilities.

A. Model with Logistic Regression: A Logistic regression binary classifier has been employed to solve the problem. This is just to have a baseline reference. Summary of performance are captured below:

- In the case of balanced data (SS20K) there is not much of gap between train and validation results.
- But in the case of unbalanced data there is a significant gap between train and validation result.

Figure 5 shows performance of logistic regression model for various subsamples of suggested dataset in term of Accuracy, F1-score, Precision, and Recall [26].

B. Model with Random Forest: Random Forest algorithm has been employed for this classification task. It's a combination of decision trees where bagging method is different bagging approach and that creates multiple models and takes the statistical average of the performance of each model is the overall performance.

In the following table model performances, in terms of accuracy and F1 score, are captured for three different datasets with itsdefault value of attributes. Without any hyper-parameter tuningit has shown improvement for precision and recall. Figure 6 shows performance of random forest model for various subsamples of suggested dataset in term of Accuracy, F1-score, Precision, and Recall.

As a follow-up, hyper-parameter like, estimator (number of decision trees) and depth of tree changed for the RF model. More optimal results are captured in Table 1. The result shows that increasing estimators does not monotonically improve the accuracy and F1 score [27]. Random Forest result is shown in table.1.

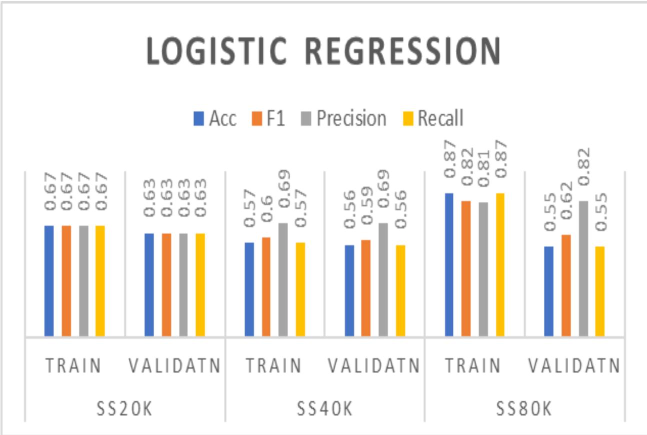


Fig. 5. Logistic regression on subsamples

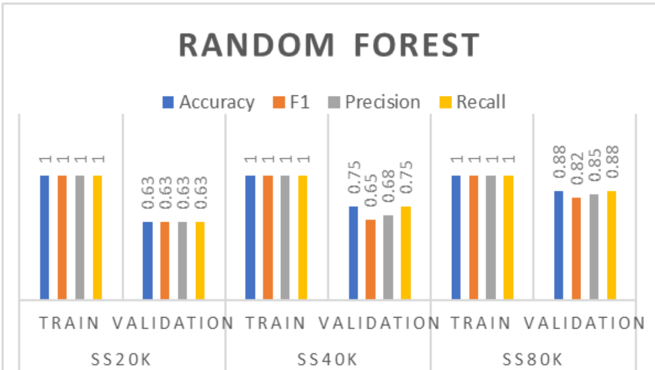


Fig. 6. Results of random forest on subsample

TABLE I. RESULT OF RANDOM FOREST AFTER HYPER-PARAMETER TUNING

Parameter and Performance Metric	**SS20k**	**SS40K**	**SS80K**
Estimators	100	200	300
Depth	12	12	12
Accuracy	0.69	0.77	0.89
F1-score	0.63	0.71	0.85

VI. Discussion

One of the most difficult and sought-after research topics in banking and finance technology is credit risk assessment. The management of consumer credit, including loans, credit cards, and the like, is also a very relevant subject for small, medium, and big institutions as well as regulatory bodies (national to international), in terms of risk identification, determination, and mitigation. Understanding customers' historical credit history, behavior, and identifying potential failure patterns are the first steps in the general approach to credit risk management. Classifying consumers from available knowledge and classifying product by response from the market,is the first and the most important step in quantifying credit risk [18]. A survey of the literature aids in understanding that this domain's problems present some of the following difficulties, which we should be aware of:

- Extremely multidimensional data.
- High imbalance is frequently caused by ground truth data that is primarily of positive class and contains relatively little negative class (yet with significant impact).
- The anonymity of meta-data because of legal requirements.

VII. CONCLUSION

As the data correctly demarcates the 'non-default' and 'default' classes, as evident in real life scenarios. In this case, data is said to be skewed with more samples falling in the positive class or 'non-default' class. As far as work is concerned which has been done in this regard up till now, the benchmark was defined by working on supervised machine learning models. In addition to experimenting with ML, this research has been taken a step further to Deep Learning and specifically to a feed forward neural network model for binary classification. This neural network model was trained using choices of optimizer and the changes of hyper-parameter to see which one performs well on the data. The outcome of the proposed work is subjected to be enhanced for several other problems in the future course of the research with the help of supervised machine learning model like neural network-based model.

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